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Automated system for construction specification review using natural language processing

Seonghyeon Moon [a,b], Gitaek Lee [a], Seokho Chi [a,b,*]

[a] Department of Civil and Environmental Engineering, Seoul National University, Seoul 08826, Republic of Korea
[b] Institute of Construction and Environmental Engineering, Seoul National University, Seoul 08826, Republic of Korea

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ABSTRACT

Existing attempts to automate construction document analysis are limited in understanding the varied semantic properties of different documents. Due to the semantic conflicts, the construction specification review process is still conducted manually in practice despite the promising performance of the existing approaches. This research aimed to develop an automated system for reviewing construction specifications by analyzing the different semantic properties using natural language processing techniques. The proposed method analyzed varied semantic properties of 56 different specifications from five different countries in terms of vocabulary, sentence structure, and the organizing styles of provisions. First, the authors developed a semantic thesaurus for construction terms including 208 word-replacement rules based on Word2Vec embedding to understand the different vocabularies. Second, the authors developed a named entity recognition model based on bi-directional long short-term memory with a conditional random field layer, which identified the required keywords from given provisions with an averaged F1 score of 0.928. Third, the authors developed a provision-pairing model based on Doc2Vec embedding, which identified the most relevant provisions with an average accuracy of 84.4%. The web-based prototype demonstrated that the proposed system can facilitate the construction specification review process by reducing the time spent, supplementing the reviewer's experience, enhancing accuracy, and achieving consistency. The results contribute to risk management in the construction industry, with practitioners being able to review construction specifications thoroughly in spite of tight schedules and few available experts.

1. Introduction

Construction specification reviews are crucial to succeeding in a construction project. A construction specification specifies every construction requirement for the project that the contractor needs to fulfill [1]. The problem is, although the construction specification is developed mainly based on national standards, several provisions include errors such as inappropriate requirements for site conditions [2]. Unless these errors are corrected, the contractors will follow the erroneous provisions in the construction phases. As a result, the project may suffer from numerous problems, such as wasted resources due to design changes, increased risks of accidents due to construction errors or unsafe installation, and conflicts or lawsuits between stakeholders due to noncompliance [3,4]. For instance, the contractors of a construction project in Qatar repeatedly undertook reconstruction works to meet the requirements specified in the construction specification. After numerous failures, the contractors finally recognized that the given requirements were inappropriate for the site conditions. In this case, the contractors suffered more than 700 non-compliance issues due to the various errors in the specification, leading to cost increases, schedule delays, and conflicts between the client and the contractor. As shown by the case, correcting the errors in the construction specification is crucial since these erroneous provisions can lead to tremendous losses for the companies involved [4,5]. Hence, both clients and contractors should review every requirement of the specification to identify erroneous provisions and appropriate solutions (e.g., change order and requirement modification) in the early phases of the project.

Despite the importance of the construction specification review in the early phases of the project, in practice, the actual review process is conducted ineffectively [6]. After several in-depth interviews with construction practitioners, it appears that the construction specification is often reviewed manually by contractors during the bidding stages. These manual reviews are vulnerable to errors and subjectivity due to the considerable volume of documents, the complicated structure of

* Corresponding author at: Department of Civil and Environmental Engineering, Seoul National University, Seoul 08826, Republic of Korea.
E-mail addresses: blank54@snu.ac.kr (S. Moon), lgt0427@snu.ac.kr (G. Lee), shchi@snu.ac.kr (S. Chi).

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provisions, the lack of time to review the specifications, and the inadequate number of experienced practitioners [7]. For example, the contractors of the construction project in Qatar mentioned above should have reviewed the construction specification, which comprised more than 4,000 pages of requirements with 13,000 references, within a two-month timeframe.

Practitioners require an automated approach to assist the construction specification review process. In these aspects, many researchers have attempted to automate the construction document analysis process using natural language processing (NLP). Although many recent studies have shown promising results in automated document analysis using NLP [8–10], there remains a gap in the research. In particular, these approaches focus on analyzing a single document and cannot recognize the varied semantic properties across different documents, such as different usages of vocabulary, different sentence structures, and different organizing styles of provisions. Having learned a homogeneous corpus composed of a single topic and a single style, the existing approaches are vulnerable to being confused when analyzing two or more documents that have different writing styles. Due to this issue, most of the existing approaches for automatic document analysis show limitations in reviewing construction specifications. In practice, reviewing construction specifications requires a minimum of two specifications: one is the client's given specification, and the other is the national standard for the construction site. During the review process, contractors should read every provision of the given specification and compare the requirements against the national standard. As every construction project provides individual specifications, the documents will obviously appear with different semantic properties. In addition, the specifications might be updated periodically throughout the project so that the review process requires analyzing different documents each time. Therefore, the authors identified a specific problem: the existing approaches cannot address the semantic conflicts across different documents, which is crucial when comparing the contents of one specification to another (e.g., the national standard). Resolving the semantic conflicts is essential to automate the construction specification review process.

This research addresses the problem of semantic conflicts existing across different documents, which impacts the automated review process of construction specifications in the field. This research aimed to develop an automated system for reviewing construction specifications with a sufficient level of industrial applicability by analyzing the varied semantic properties using NLP. To address the practical limitations of existing approaches—in which they cannot recognize the varied semantic properties across different documents—the authors developed machine learning-based NLP models to understand specifications with a different vocabulary, different sentence structures, and differently organized provisions. In addition, to make it easier for field engineers to use the system on-site, the authors designed a web-based prototype for the automated construction specification review system. First, the authors proposed a semantic thesaurus for construction terms based on the Word2Vec embedding model and the *PivotTermDetermination* algorithm to understand different vocabulary. Second, the authors developed a named entity recognition (NER) model that recognizes requirement keywords from natural language sentences based on Word2Vec embedding and the bi-directional long short-term memory (Bi-LSTM) architecture with a conditional random field (CRF) layer. Third, the authors developed a provision-pairing model that identifies the most relevant provision for every given provision based on Doc2Vec embedding and semantic similarity calculations. Lastly, the authors designed a user interface (UI) and user experience (UX) for the web-based prototype for the automated construction specification review. The web-based prototype supports access by field engineers regardless of spatial constraints and provides an easy UX while reviewing the specifications.

The result of this research is an automated system for reviewing construction specifications. Once the user selects the target provision (i.e., required to be reviewed) from the construction specification and selects other specifications to be compared, the system automatically provides the most relevant provisions from the selected specifications and compares the qualitative requirements between the two provisions. While the system can also recommend specifications that might be reliable for comparison, the process of specification recommendation is beyond this research scope.

Every construction specification analyzed in this research was written in English and related to functional areas of road construction. English specifications were selected since the English language is the most commonly used in international construction projects. However, the proposed method can analyze specifications written in other languages because the language model learns the distributed information of input text instead of an alphabet or grammar convention. In addition, as the authors defined information categories necessary for project risk management in common cases, every specification can be parsed irrespective of the construction items or functional areas.

2. Literature review

2.1. Construction specification review

The authors examined a construction specification and demonstrated the difficulty of the manual review process. Because of the data availability issue that specifications are commonly confidential, the authors could only investigate a construction specification of a road construction project in Qatar from 2014. Since Qatar does not have a national standard specification for road construction, the contractors reviewed the construction specification by comparing the requirements to those of the referred national standards from other regions. However, the construction specification cited national standards from diverse origins indiscriminately, which impacted the contractors' ability to review the provisions' appropriateness to the site condition.

The authors counted every reference mentioned in the construction specification. The references amounted to 12,995, including the United Kingdom (UK) for 5,024 (39%), the European Union for 3,765 (30%), the United States for 2,491 (19%), the international standards for 1,196 (9%), and other sources for 519 (4%) references. It seemed logical that the UK was referred to the most as the UK and Qatar are historical partners; however, some of the provisions were highly inappropriate to the site environment. For example, although Qatar has always exhibited a desert climate, with construction sites being hot and dry, the provisions specifying the temperature of asphalt mixtures referred to the UK's national standard, which presents a cold climate.

In addition to the indiscriminate references, the large volume and complicated contents of the construction specification also impacted the review. The construction specification contained 4,790 pages, 29 major chapters (e.g., "Concrete and Roadworks"), and 285 subchapters (e.g., "Concrete Road Pavements, Concrete Plants, and Curing"). At the chapter level, the categories were not mutually exclusive; for instance, the "Concrete Road Pavements" subchapter was located under the "Roadworks" chapter rather than the "Concrete" chapter. At the provision level, most sentences were too long to be understood—over four to five lines without a period—and the same construction elements or objects (e.g., coarse aggregate, curing temperature, and tack coat) sometimes followed different references. Thus, contractors found it challenging to review, understand, and analyze all the contents of the specification manually, with the risk of overlooking errors or crucial issues and misinterpreting the provisions; this is likely to result in unexpected problems during actual construction in the field.

In reality, in the Qatar example, the contractors failed to detect erroneous provisions at the bidding stages due to the tight schedule, an insufficient number of available professionals, and the large volume of information. Consequently, erroneous provisions eventually led to construction errors, which caused compensation to the client that was almost equivalent to the project's initial cost.

2.2. Automated document analysis in the construction industry

Many researchers have attempted to analyze construction documents automatically. Early studies focused on classifying or clustering construction documents for efficient management by practitioners [11–18]. These approaches were greeted positively by the construction industry, reducing the considerable amount of resources needed to manage the knowledge in the documents [19]. However, as construction projects become larger and more complex, these approaches to document-level analysis cannot provide users with practical information. In practice, construction practitioners require information of a more detailed level (i.e., words, sentences, or paragraphs); for example, instead of determining whether the document type is a contract document or an accident report, practitioners need to know which paragraphs include high risks for the construction project.

Building on the recent advances in data storage, computer processing, and deep learning methodologies, NLP in the construction industry has also entered a new phase. A literature review shows that the number of academic studies has experienced an explosive increase in analyzing construction documents at more detailed levels. Several studies estimated the economic feasibility and risks of construction projects by analyzing the contract documents in the bidding phases [20–25]. Numerous research groups have worked on automatic compliance checking, which is a process of reviewing the documents to determine whether the provisions comply with the regulatory requirements [3,4,9,10,26–31]. In recent papers, the authors discovered that a significant proportion of the automatic compliance checking systems had been automated with promising performance results [3,32,33]. When accidents occur on a construction site, it is mandatory for contractors to investigate the causes and write accident reports. Analyzing these accident reports to identify risk factors and provide alternatives for prevention is one of the most popular research areas [6,8,34–39]. In addition, plenty of studies have attempted to analyze facility inspection reports in order to extract damage occurrence mechanisms, estimate the damage occurrence and scale, and predict the costs of repair and rehabilitation [40].

However, despite a number of existing studies on automated construction document analysis, practitioners in the field still conduct the specification review process manually. They use optical character recognition programs to detect text sentences from specifications and search keywords using search functions that the commercial text editor programs provide (i.e., the shortcut of F3). The most critical reason that the approaches of existing studies are difficult to apply to the construction site is already described in the introduction section: existing

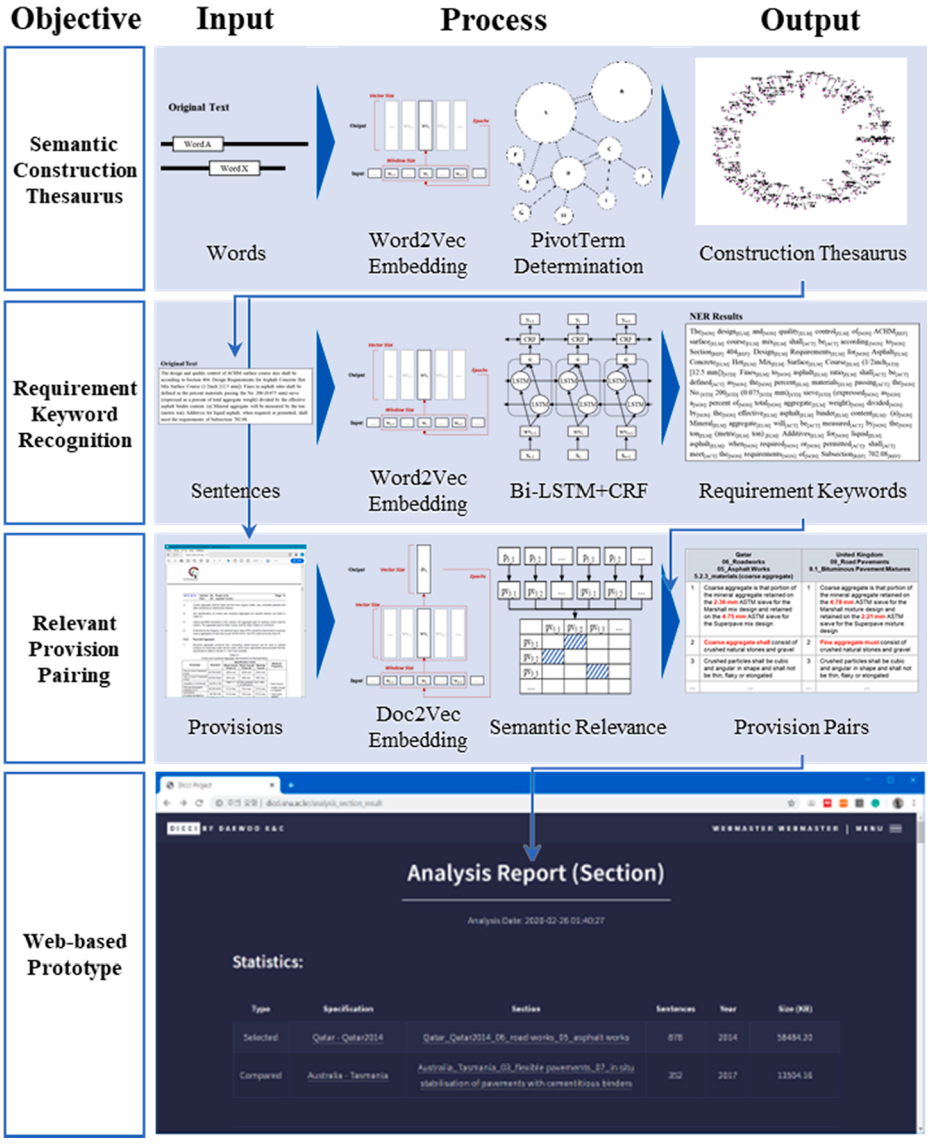


Fig. 1. Research framework.

approaches focus on analyzing the semantic features in a single type of document—they cannot understand the varied semantic properties between different documents. As noted earlier, the construction specification review requires a minimum of two documents, which indicates that automated review systems should understand the varied semantic properties of different documents. The authors attempted to fill the research gap via this research.

3. Model development

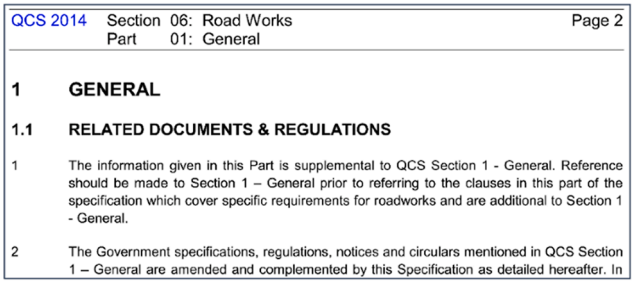
The objective of this research was to develop an automated system for reviewing construction specifications by analyzing the different semantic properties using NLP techniques. The overall research framework is provided in Fig. 1. The research included four specific objectives. The first specific objective was to develop a semantic construction thesaurus to facilitate the computer's understanding of the text irrespective of the different vocabulary used. The varied vocabulary of the different documents causes a misunderstanding of provisions during the text comparison task for automated specification reviews. The authors addressed this problem by developing a domain-specific thesaurus of the construction industry. The second specific objective was to develop a requirement keyword recognition model to facilitate the computer to understand the text irrespective of the sentence structure. The different sentence structures obstruct the comparison of two provisions from different documents during automated specification reviews. The authors addressed this problem with a machine learning-based NER model that can recognize construction keywords from sentences. The third specific objective was to develop a provision-pairing model to facilitate the computer to understand the text irrespective of differently organized provisions, which obscure the text comparison process during specification reviews. The authors addressed this problem by pairing the relevant provisions based on their semantic relevance. The last specific objective was to design the UI and UX of the web-based prototype of the automated system for reviewing construction specifications. Using the review functions developed by the research, the prototype facilitates easy usage of the system by field engineers.

3.1. Data preparation

A total of 56 construction specifications were collected for the research, which originated in Australia, Canada, Qatar, the UK, and the United States. Detailed information on the collected specifications can be found in previous work by the authors [41]. As the collected specifications were in PDF format, the computer could not modify or analyze the contents. Therefore, every PDF file was converted into TXT format, which allows the text to be modified [25]. As the quality of the data had an immediate effect on the analysis results, the conversion processes were performed manually; that is, the drag-copy-paste approach was used to extract every sentence from the PDF file one at a time (Fig. 2). As a result, a total of 2,507 provisions (i.e., 19,338 sentences) were prepared for analysis.

Text preprocessing is one of the essential steps in NLP; the purpose of this step is to ensure the quality of the text data. Preprocessing is commonly conducted using several techniques, including tokenization, stopword removal, stemming, and lemmatization [42]. First, every sentence of the prepared text data was tokenized based on the multi-gram (i.e., n-grams) approach, counting neighbor terms that frequently appear in the text. In order to consider the construction-related phrases composed of several general terms, the authors determined the size of n as being two to five terms. For example, "job" and "mix," which are commonly used in daily life, were linked to each other and interpreted as "job mix" (i.e., a mixture of aggregate and asphalt binder) by the language model. Second, the stopwords within the tokenized sentences were removed based on the predetermined stopword list. The authors carefully customized the stopword list for a better representation of text data. The stopword list included "a," "be," "is," "are," "was," "were," "the," "this," "these," "that," "those," and "of." Third, lemmatization was selected for the analysis rather than stemming as lemmatized words showed an adequate level of information while the stemmed words diluted the semantic information of provisions.

Original Specification (.PDF)



QCS 2014 Section 06: Road Works Page 2
Part 01: General

1 GENERAL

1.1 RELATED DOCUMENTS & REGULATIONS

1 The information given in this Part is supplemental to QCS Section 1 - General. Reference should be made to Section 1 – General prior to referring to the clauses in this part of the specification which cover specific requirements for roadworks and are additional to Section 1 - General.

2 The Government specifications, regulations, notices and circulars mentioned in QCS Section 1 – General are amended and complemented by this Specification as detailed hereafter. In

(a)

Manually Extracted Sentences (.TXT)

qatar 06 roadworks 01 general 01 related documents & regulations
The information given in this Part is supplemental to QCS Section 1 - General
Reference should be made to Section 1 - General prior to referring to the clauses in this part of the specification which cover specific requirements for roadworks and are additional to Section 1 - General
The Government specifications, regulations, notices and circulars mentioned in QCS Section 1 - General are amended and complemented by this Specification as detailed hereafter
In the case of any ambiguity or discrepancy the provisions of this Specification shall prevail over the provisions of the aforementioned Government published specifications

(b)

Fig. 2. Example of data format conversion: (a) original PDF, (b) manually extracted TXT.

3.2. Semantic construction thesaurus

A thesaurus is a dictionary that describes the relationship between terms, such as a synonym, hypernym, and hyponym [43–46]. A thesaurus is also known as a "term map" because it replaces one term with another term so that the language model can comprehend the terms identically [47]. For example, the relationship "the terms 'asphalt' and 'bituminous' are similarly used in the corpus" would replace every "asphalt" with "bituminous" in the given text.

A thesaurus is commonly built by industrial professionals who have a profound technical knowledge of a specific domain or by using existing synonym dictionaries. However, these approaches might be restricted to prevalent and refined information. The authors advanced the construction thesaurus based on Word2Vec; hence, the thesaurus was developed automatically to consider new vocabulary from new documents. The Word2Vec model analyzed the usage patterns of words (i.e., word distribution in the sentence) and determined the semantic relationships of similarly used words [48]. Among the two architectures of Word2Vec (Continuous Bag of Words (CBOW) and skip-gram), the authors applied the skip-gram architecture for text embedding according to the developer's note: CBOW is faster while skip-gram infers text better [49]. The following figure illustrates a conceptual framework of the skip-gram architecture (Fig. 3). First, the Word2Vec model initializes every word vector with random values. Then, using a single word vector as an input, the Word2Vec model tries to predict the neighbor word vectors. The Word2Vec model adjusts the values of the input word vector to decrease the prediction errors. Repeating the prediction and adjustment process for every word, the Word2Vec model can train the word distributions. The results are the word vectors, of which values would be akin to the similarly used words in other sentences.

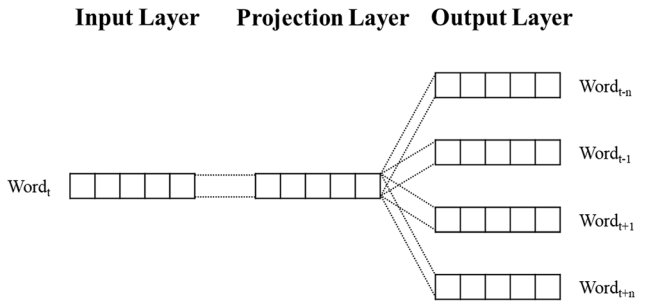


Fig. 3. Skip-gram architecture.

The Word2Vec model trained 346,950 words (8,692 unique terms) from the 19,338 sentences. The hyperparameters were determined based on the grid search conducted by the authors (Table 1). "VectorSize" (i.e., vector size) implies the dimension of word vectors. "WindowSize" (i.e., window size) indicates the number of surrounding words that are considered to learn the usage pattern of a word. Too rarely occurring words, of which the frequency was less than the value of "MinimumCount" (i.e., minimum count), were discounted during the training. "Epochs" represent how many times the model trains the whole data repeatedly.

A similar word dictionary was developed based on the cosine similarity between the word vectors. The authors calculated the cosine similarity between the word vectors and dropped some words where the similarity showed less than 0.5. Finally, the similar-word dictionary included pairs of similar words, which facilitates the text analysis to replace a word with its most similar word. However, the thesaurus might frequently hesitate to determine which word to change and which to keep; that is, a problem occurs where several words are converted to each other recursively. For example, the words "tyr," "tir," and "pneum" (i.e., the lemmas of "tyre," "tire," and "pneumatic," respectively) turned out to be forming a recursive network (Fig. 4).

The existing works about developing thesaurus did not focus on the recursive replacement issue since the relationships between words included the relevance between words as well as which word is more important. The relevant importance between words was naturally investigated when the domain professionals developed the thesaurus [29,35]. However, these manually investigated relationships between words did not suit the automated construction review process due to the biased results of the domain professionals' judgment. To solve this issue, the authors proposed an algorithm (i.e., *PivotTermDetermination*) to identify the most important word from a word group that shows a recursive replacement problem. When developing the *PivotTermDetermination* algorithm, the PageRank algorithm that identifies the importance of web pages based on hyperlink relationships inspired the authors [50]. For two or more words that are similarly used, the most important word should be identified; the thesaurus would, therefore, replace the other words with the most important word. Since the most important word is not changed to any other word in the text, we named it "pivot term." In other words, the thesaurus recognizes the pivot term

Table 1
Hyperparameters of the Word2Vec model.

Hyperparameter	Range of grid search (Start, end, step)	Value	Description
VectorSize	(50, 500, 50)	200	The dimension of word vector
WindowSize	(3, 20, 1)	10	The number of adjacent words to learn the word distribution
MinimumCount	(0, 30, 1)	10	The minimum frequency of each word to learn the distribution
Epochs	(50, 500, 50)	200	The total number of repeats to learn the whole training data

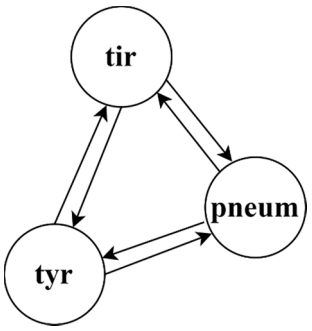


Fig. 4. Sample of recursive word replacement.

as being central when handling the relevant words within the sentence. The authors proposed the concept of inflow and outflow for word replacements. The inflow of a word indicates the number of other words that have the current word as a similar word. In reverse, the outflow of a word indicates the number of words that it showed to be similar to. However, such words of which the similarity score is less than 0.5 were ignored in the research. Similar to the PageRank algorithm (where a web page that receives hyperlinks from many other web pages would be regarded as important), where a word has a high inflow and a small outflow, it would be regarded as an important word (i.e., a pivot term) [51]. In other words, the pivot terms can be determined based on the number of links and the flow margin. The flow margin is calculated in Eqs. (1) to (4):

$$g(w', W_w) = \frac{n(W_w) - index(w'|sorted(W_w))}{n(W_w)} \tag{1}$$

$$inflow_w = \sum_{w' \in W_{w,IN}} g(w', W_{w,IN}) * c(w, w') \tag{2}$$

$$outflow_w = \sum_{w' \in W_{w,OUT}} g(w', W_{w,OUT}) * c(w, w') \tag{3}$$

$$f_w = inflow_w - outflow_w \tag{4}$$

where w and w' indicate an individual word, and W indicates a set of words. In particular, $W_{w,IN}$ indicates a set of words that includes the word w as one of its most similar words and $W_{w,OUT}$ indicates the set of the most similar words of the word. $g(w', W_w)$ is a weight function of w' from the list of sorted elements of W_w based on the cosine similarity to w. f_w means the flow margin of a word, and the function $c(w, w')$ returns the cosine similarity between the input word vectors. The algorithm of determining the pivot term (i.e., *PivotTermDetermination*) is provided as:

(1) If the flow margin of a word is positive, keep the word.
(2) If the flow margin is negative, replace the word with other words:
 (2-1) If the flow margin of the most similar word is positive, replace the original word with the most similar word.
 (2-2) Else, do (2-1) with the next most similar word.
(3) If there is no word for which the flow margin is positive, replace the original word with the most similar word, and do (1).

3.3. Requirement keyword recognition

NER is a text-mining technique used to recognize informative keywords from text [52–54]. Since the informative keywords have predetermined categories, they are so-called "named entities." For example, if a NER model learns the 'reference' categories and reads the sentence "The samples should be tested according to ASTM D6927," the model will recognize "ASTM D6927" as a named entity of 'reference.'

NER models can be developed using two approaches: rule-based and machine learning-based. The rule-based approaches conduct NER by

building syntactic or semantic rules with morphological analysis, whereas the machine learning-based approaches train the model with text data labeled with predetermined categories. The rule-based approaches are known for their supreme accuracy and ability to interpret the results; however, they are limited to the built rules from the training data and are difficult to apply to different sentence structures [30,35,55]. Due to its robustness, expandability, and the fewer required resources for model training, the machine learning-based NER has been used in recent studies [9,56].

The authors developed a deep learning model for NER to recognize the requirement keywords based on Bi-LSTM architecture with a CRF layer. The Bi-LSTM + CRF is one of the most recent and powerful architectures in NLP, especially for NER tasks [41,57]. Because of the following issues, the Bi-LSTM + CRF architecture was seemed suitable for requirement keyword recognition from provisions. First, the construction specifications commonly include more complex and longer sentences than other documents. The Bi-LSTM structure addresses this issue with selective remembrance of informative keywords—even if the keywords are situated far from each other in the sentence [58]. Meanwhile, the same words might represent different information according to the neighbor words in the text. For example, the word "mix" can be used as a verb to explain some actions or as a noun to explain an object, like "job mix formula." The CRF structure addresses the issue of reasoning the appropriate information category of each word by considering the outputs of neighbor words [59–61].

In detail, the NER model has five layers: input layer, backward LSTM layer, forward LSTM layer, activation layer, and CRF layer (Fig. 5). At the input layer, the model utilizes the word vectors as inputs generated from the Word2Vec model developed in Section 3.2 'Semantic construction thesaurus.' The backward and forward layers consist of LSTM units for every input word vector; the model thereby can learn the bi-directional sequences of surrounding words. The activation layer normalizes the output shape between 0 and 1 based on the ReLu (i.e., Rectified Linear Unit) activation function. The CRF layer adjusts the output according to the surrounding output values.

The hyperparameters of the NER model were determined based on the grid search conducted by the authors (Table 2). "MaxSentLen" (i.e., maximum sentence length) indicates the maximum number of words in each input sentence, representing the NER model's input size. "LSTM Units" and "Dense Units" indicate the dimensions of the Bi-LSTM and dense units, respectively, which represent the dimension of hidden layers. "DropOut" (i.e., drop out ratio) indicates the ratio of thinning weights during the training process; hence, the model avoids overfitting the training data. "BatchSize" (i.e., batch size) indicates the number of samples that the model learns during one iteration, and "Epochs" represents how many times the model trains the whole data repeatedly.

Table 2
Hyperparameters of the NER model.

Hyperparameter	Range of grid search (Start, end, step)	Value	Description
MaxSentLen	(30, 50, 10)	50	The maximum number of words in each input sentence
LSTM Units	(256, 1,024, 256)	1,024	The dimension of Bi-LSTM units
Dense Units	(50, 200, 50)	50	The dimension of the dense matrix
DropOut	(0, 0.5, 0.1)	0.2	The ratio of thinning weights during the training process
BatchSize	(16, 64, 16)	32	The number of samples to train during one iteration
Epochs	(50, 500, 50)	200	The total number of repeats to learn the whole training data

According to in-depth interviews with 25 practitioners in the construction industry, five information types were identified that should be recognized from construction specifications for project risk management. Practitioners' personal information is provided in Table 3. The information types (i.e., the categories of words which the NER model would recognize) included people and organization in charge (organization: ORG), activities required (action: ACT), construction and installation items (element: ELM), quality standards and criteria (standard: STD), and relevant references (reference: REF).

Six construction practitioners were involved in assigning each word to its proper category, and, finally, 4659 sentences were prepared for NER model training. To train the model with various sentence structures, the authors prepared data by randomly selecting sentences from the 56 construction specifications. In addition, the authors randomly divided the data into a training set and validation set to avoid the problem of overfitting. Specifically, 70% of labeled sentences (3261 sentences) were used to train the model, and the remaining 30% (1398

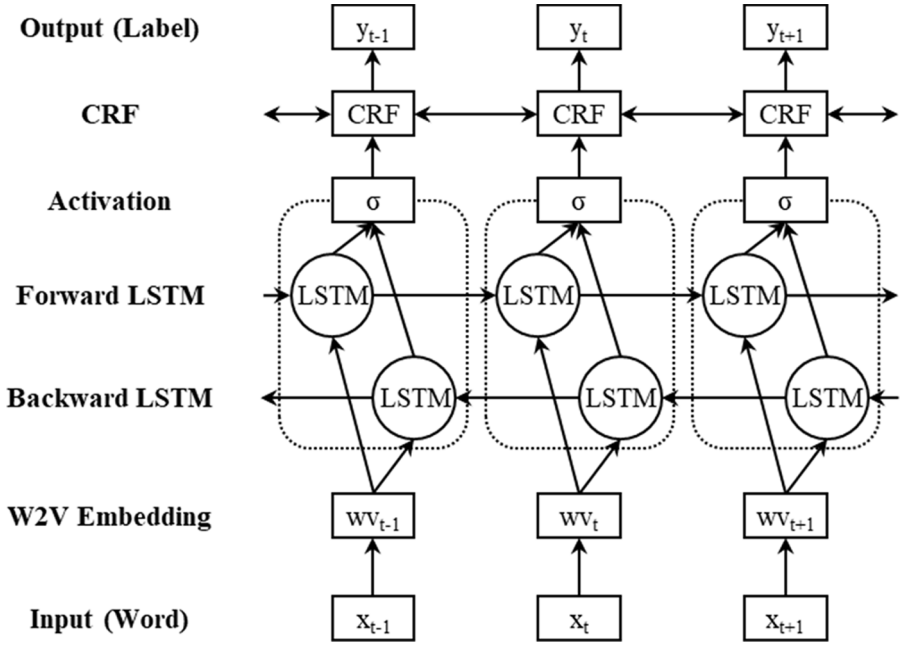


Fig. 5. NER model framework.

Table 3
Personal information of interviewees.

ID	Department	Position	Work Experience (Year)
1	Design	Director	30
2	Design	Director	30
3	Engineering	Department Head	20
4	Engineering	Department Head	20
5	Engineering	Department Head	20
6	Design	Department Head	15
7	Design	Department Head	15
8	Engineering	Deputy Department Head	15
9	Engineering	Deputy Department Head	15
10	Engineering	Deputy Department Head	15
11	Engineering	Deputy Department Head	15
12	Engineering	Deputy Department Head	15
13	Engineering	Manager	10
14	Engineering	Manager	10
15	Engineering	Manager	8
16	Design	Manager	6
17	Design	Researcher	6
18	Design	Researcher	5
19	Engineering	Researcher	5
20	Engineering	Researcher	5
21	Design	Researcher	4
22	Engineering	Researcher	4
23	Engineering	Researcher	3
24	Engineering	Researcher	3
25	Engineering	Researcher	3

sentences) were used to validate the model. More detailed information can be found in previous work by the authors [41]. To extend the previous work, the authors integrated the inputs and outputs of the NER model with the current research. The semantic construction thesaurus was applied to the inputs to understand the text better, while the outputs were utilized to identify the differences between provisions.

3.4. Relevant provision pairing

Semantic relevance is a measure of how similar the subjects of the two texts are. Identifying the most relevant provision is crucial to the automated specification review, as the requirements should be reviewed within the same subject area. Many researchers have attempted to develop similar-case retrieval systems that identify the most similar text (i.e., document or sentence) [15,34]. Despite their promising performance, these systems are restricted to analyzing the text at the document or sentence level. This limitation is critical for the specification review process because of the following reasons. First, since document-level text retains highly varied information, the user should still investigate every sentence to find the most relevant requirement even if the most relevant document was provided. On the other hand, since sentence-level text provides too specific information, the review process would struggle to figure out which construction item the sentence describes. In order to address these issues, the authors suggested analyzing the text at the provision-level, which seems to retain the proper amount of information for text comparison. A provision comprises a set of continuous sentences that share the lowest level of heading. As mentioned in the introduction section, the granularity of the provisions might vary across different documents due to the organizing styles of provisions. For example, small provisions may have only one or two sentences, while heavy provisions might include dozens of sentences. However, since the sentences in a provision correspond to a unique topic in the document, the model (or the reviewer) can understand the contents and requirements without confusion.

The provision-pairing model developed in this research provides the most relevant provision based on semantic relevance. The authors embedded the provision text based on Doc2Vec. Doc2Vec is a machine learning-based text embedding technique that represents longer text (i.e., sentence, paragraph, or document) in a dense numeric vector [62–64]. Similar to the Word2Vec model, the Doc2Vec model also learns the distributed representation of words within every text. Since the provisions from different specifications are organized differently, comprehension of word distributions is invaluable for understanding the semantic information of each provision. Among the two architectures of Doc2Vec—the Paragraph Vector with Distributed Memory (PV-DM) and the Paragraph Vector with Distributed Bag of Words (PV-DBOW)—the authors applied the PV-DBOW architecture for provision embedding. The PV-DBOW architecture is known to be faster on learning and predicting the document vector better on short-input text (i.e., a few dozens of words) [63].

The Doc2Vec model trained 2,507 provisions. The hyperparameters were determined based on the grid search conducted by the authors (Table 4). "VectorSize" (i.e., vector size) implies the dimension of the provision vectors. Other hyperparameters (e.g., "WindowSize," "MinimumCount," and "Epochs") share the description in the Word2Vec model explained in Section 4.1 Semantic Construction Thesaurus.

The authors calculated the semantic relevance between provisions based on the cosine similarity between the provision vectors. Cosine similarity is calculated as the inner value of the angle between two vectors [65], commonly used in NLP to represent the text similarity. In this research, the authors modified the cosine similarity by limiting the results between 0 and 1; hence, the value represents the semantic relevance more intuitively (Eq. (5)).

$$Semantic\ \ Relevance = \frac{(Cosine\ \ Similarity + 1)}{2} \tag{5}$$

To verify the provision pairing results, case studies were conducted with eight construction experts whose average work experience was 23 years. They selected a specification for a highway construction project in Qatar in 2014 as an analysis target and five other specifications as comparison targets. In particular, the expert group focused on provisions related to asphalt pavement which they were versed in. Section 5 ('Asphalt Work') of Chapter 6 ('Road Works') from the Qatar specification (i.e., Qatar_2014_06_05) was chosen to be analyzed, and ten sections from the other specifications were also selected. The experts acknowledged that these sections were relevant to 'Qatar_2014_06_05' (Table 5).

The analysis target section 'Qatar_2014_06_05' had 75 provisions. For every comparison section, the provision-pairing model returned the most relevant provision from the compared section for each provision in 'Qatar_2014_06_05.' The pairing threshold for semantic relevance was determined as 0.8 based on several preliminary experiments. In other words, a provision that had no provision for which the semantic relevance was more than 0.8 from the compared section was returned as 'No Relevant Provision.'

4. Results

4.1. Results of the semantic construction thesaurus

Omitting too rare terms, for which the frequency was less than ten

Table 4
Hyperparameters of the Doc2Vec model.

Hyperparameter	Range of grid search (Start, end, step)	Value	Description
VectorSize	(50, 500, 50)	500	The dimension of the provision vector
WindowSize	(3, 20, 1)	10	The number of adjacent words used to learn the text distribution
MinimumCount	(0, 30, 1)	30	The minimum frequency of each word to learn the distribution
Epochs	(50, 500, 50)	200	The total number of repeats to learn the whole training data

Table 5
Comparison sections for verifying provision pairing.

Country	Regulatory Authority	Last Edited	Section Title	Tag
Australia	Tasmania	2017	Warm Mix Asphalt	Australia_Tasmania_2017_04_09
Australia	Tasmania	2017	Bitumen Crumb Rubber Asphalt	Australia_Tasmania_2017_04_21
Qatar	–	2010	Unbound Pavement Materials	Qatar_2010_06_04
Qatar	–	2010	Asphalt Work	Qatar_2010_06_05
United States	Alabama	2018	Planning (Milling) of Existing Pavement	United States_Alabama_2018_04_08
United States	Arkansas	2014	Asphalt Concrete Hot Mix Base Course	United States_Arkansas_2014_04_05
United States	Arkansas	2014	Asphalt Concrete Hot Mix Binder Course	United States_Arkansas_2014_04_06
United States	Arkansas	2014	Asphalt Concrete Hot Mix Surface Course	United States_Arkansas_2014_04_07
United States	Arkansas	2014	Asphalt Concrete Patching for Maintenance of Traffic	United States_Arkansas_2014_04_14
United States	Connecticut	2018	Bituminous Concrete Materials	United States_Connecticut_2018_04_19

based on the "Minimum Count," the Word2Vec model returned a unique vector for 4,049 terms. Then, the cosine similarities between word vectors were calculated, and the five most similar terms with a similarity larger than 0.5 were listed for each term to build a similar-word dictionary. As a result, the authors gathered 374 records. The records of the similar-word dictionary are illustrated in a word network, with the directions 'from the original term to similar terms' (Fig. 6).

The semantic construction thesaurus was developed by analyzing every link between the terms and the similar words based on a simple concept of the flow margin provided in Section 3.2 'Semantic construction thesaurus.' The word replacement rules (i.e., each term to the pivot) were illustrated in the form of a word network (see Fig. 7). The thesaurus rules will replace every term at the start points with the endpoint throughout the text.

As the semantic construction thesaurus has no correct answers, the authors evaluated the results by investigating several randomly selected rules. Table 6 provides the samples of the semantic construction thesaurus. The records with indexes 1, 2, 3, and 5 turned out to be reasonable. Removing a few rules that the thesaurus misinterpreted (e.g., the index number of 4), the authors finalized the semantic construction thesaurus with 208 replacement rules.

4.2. Results of requirement keyword recognition

The authors evaluated the NER model in terms of precision, recall, and F1 score, which are the most widely used indicators for classification performance measurement. Given a sentence, the NER model returned the most probable category (i.e., ORG, ACT, ELM, STD, REF,

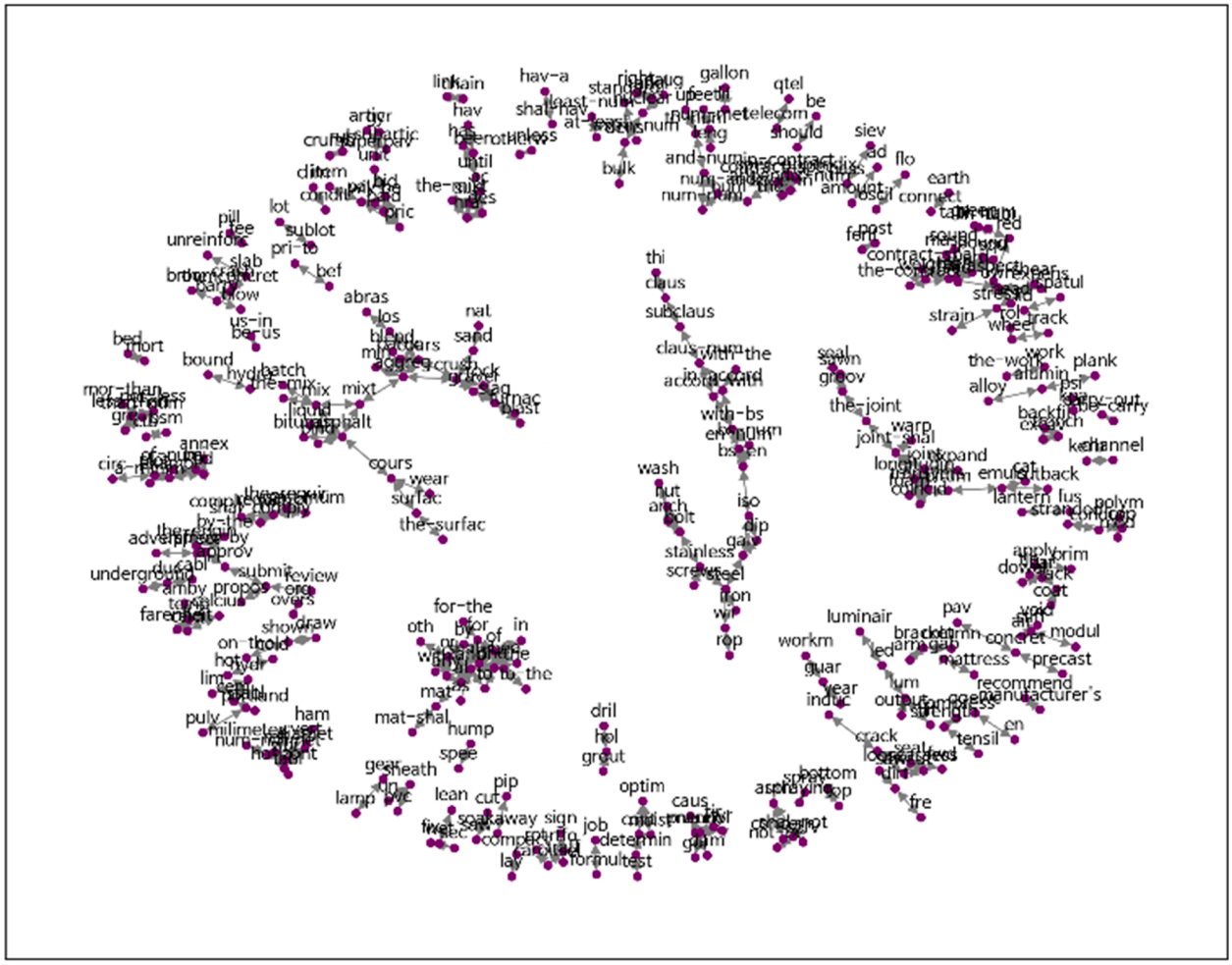


Fig. 6. Word network for the similar-word dictionary.

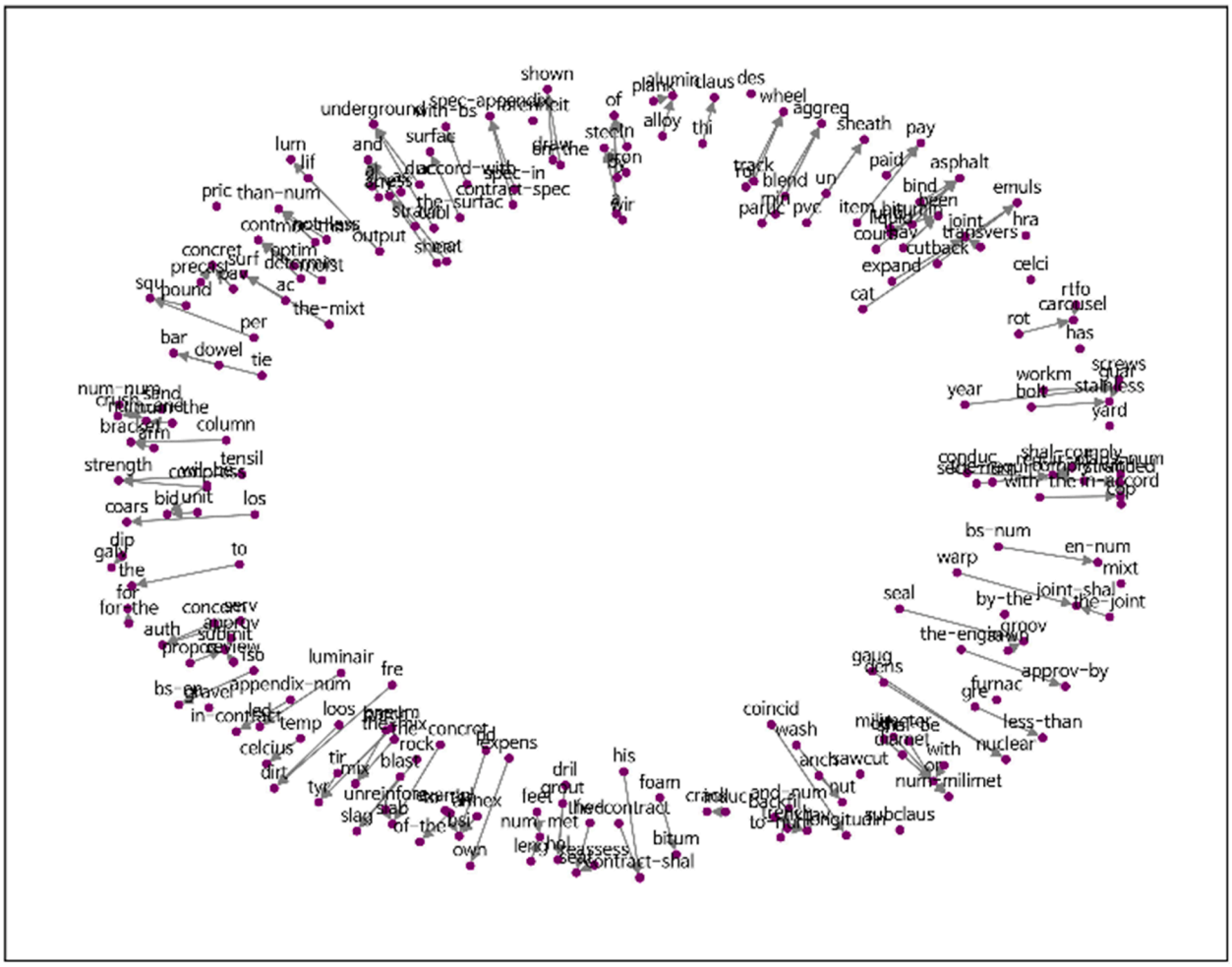


Fig. 7. Word network for semantic construction thesaurus.

Table 6
Samples of word replacement results.

Index	Target Term		Pivot Term	
	Original Token	Lemma	Original Token	Lemma
1	temperature	temp	Celsius	Celsius
2	tire	tir	tyre	tyr
3	propose	propos	submit	submit
4	approval	approv	submit	submit
5	bituminous	bitumen	asphalt	asphalt

and NON) for every word in the sentence. The classification results for each word category are provided in the confusion matrix (Table 7). Consequently, the NER model showed promising performances for each word category, with a total F1 score of 0.928 (Table 8).

Table 8
NER performance for each word category.

	Word Categories					
	NON	ORG	ACT	ELM	STD	REF
Precision	0.901	0.950	0.901	0.907	0.899	0.958
Recall	0.910	0.927	0.912	0.919	0.891	0.919
F1 Score	0.905	0.939	0.906	0.913	0.895	0.938

An example of the NER results is illustrated in Fig. 8. The NER model assigned a category to every word in the input sentences. For detailed information about the requirements for keyword recognition, the readers are recommended to see the authors' previous work [41].

Table 7
Confusion matrix of requirement keyword recognition.

		Actual Categories						
		NON	ORG	ACT	ELM	STD	REF	TOTAL
Predicted Categories	**NON**	10,428	14	407	660	57	12	11,578
	ORG	16	588	0	30	0	0	634
	ACT	321	2	4,493	90	23	0	4,929
	ELM	607	15	66	9,483	122	30	10,323
	STD	73	0	19	132	1,825	0	2,049
	REF	20	0	1	60	3	952	1,036
	TOTAL	11,465	619	4,986	10,455	2,030	994	30,549

The$_{[NON]}$ design$_{[ELM]}$ and$_{[NON]}$ quality$_{[ELM]}$ control$_{[ELM]}$ of$_{[NON]}$ ACHM$_{[REF]}$ surface$_{[ELM]}$ course$_{[ELM]}$ mix$_{[ELM]}$ shall$_{[ACT]}$ be$_{[ACT]}$ according$_{[NON]}$ to$_{[NON]}$ Section$_{[REF]}$ 404$_{[REF]}$. Design$_{[ELM]}$ Requirements$_{[ELM]}$ for$_{[NON]}$ Asphalt$_{[ELM]}$ Concrete$_{[ELM]}$ Hot$_{[ELM]}$ Mix$_{[ELM]}$ Surface$_{[ELM]}$ Course$_{[ELM]}$ (1/2inch$_{[STD]}$ [12.5 mm])$_{[STD]}$.

Fig. 8. Example result of requirement keyword recognition.

4.3. Results of relevant provision pairing

As mentioned in Section 3.4 'Relevant provision pairing,' the provision-pairing model parsed the sections of Table 5 to individual provisions and paired the most relevant provision for every provision of 'Qatar_2014_06_05.' With the semantic relevance threshold of 0.8, the experiment results showed a fine performance of 84.40% accuracy (i.e., (74 + 559)/750). In other words, during sifting through the most relevant provisions for the total of 75 provisions of 'Qatar_2014_06_05' with the ten other sections (i.e., a total of 750 pairing cases), the pairing model reported 74 cases of true positive and 559 cases of true negative (Table 9). In detail, the F1 score for identifying a relevant provision was 0.558, whereas the F1 score for dropping a non-relevant provision was 0.905 (Table 10). These performance metrics indicate a considerable improvement is essential for identifying relevant provisions. The authors plan to collect a larger amount of qualified training data, conduct hyperparameter optimization, and conduct sensitivity analysis for the threshold of semantic relevance. However, the provision-pairing model could, for the moment, contribute to the construction specification review process. In particular, the model eliminates non-relevant provisions quickly and thus noticeably reduces the review time.

An example of a provision pair is illustrated in Fig. 9. The model analyzed the provision for delivery, storage, and handling from 'Qatar_2014_06_05' and paired the most relevant provision from 'United States_Connecticut_2018_19_04.'

4.4. Web-based prototype

The authors designed the UI and UX of a web-based prototype of the automated system for reviewing construction specifications. The prototype was developed via Django, an open-sourced web application framework based on Python. For easy usage by field engineers, the prototype provides several functions developed based on the research, including data management, relevant provision recommendation, requirement keyword detection, and a print-out of the review report.

First, the user can select a specification for automated review. As it is linked to a pre-built database, the prototype supports any cross-review between 56 construction specifications. The user can select a specification to review on the left side of the page and another specification to compare on the right side (Fig. 10(a)). Then, the prototype recommends the five most relevant provisions for every provision of the selected specification (Fig. 10(b)). The provision-pairing model is applied to the recommendation system. Once the user selects one of the listed provisions, the prototype reviews the pair of provisions automatically. If a relevant provision is absent, the page returns the advice 'No Relevant Provision.' After identifying a pair of provisions, the prototype detects the requirement keywords and compares the keywords between the provisions (Fig. 10(c)). The NER model is applied to the keyword detection process. Hereby, the user can identify the differences between the requirements at a glance. The last step is to export the specification review results in report format (Fig. 10(d)). Note that the prototype provides automated options for every step of the review process. The review results for every provision of the selected specification are listed with a notification: whether the requirements are the same or different from the compared specification. As such, practitioners can easily pick out the provisions where the requirements appear to be different from the compared specification (i.e., require in-depth attention).

Table 9
Confusion matrix of provision pairing.

		Actual Pairs		
		Relevant	No Relevant	Total
Predicted Pairs	**Relevant provision**	74	61	135
	No relevant provision	56	559	615
	Total	130	620	750

Table 10
Provision-pairing performance.

	Provision Pairs	
	Relevant provision	No relevant provision
Precision	0.548	0.909
Recall	0.569	0.902
F1 Score	0.558	0.905

5.2.9 Delivery, Storage and Handling

1 Materials shall be so stored and handled as to assure the preservation of their quality and fitness for use. Materials, even though approved before storage or handling, may again be inspected and tested before use in the Works.

2 Stored material shall be located so as to facilitate their prompt inspection. All storage locations on land not owned by the Contractor shall be restored to their original condition at the Contractor's expense.

3 Handling and stockpiling of aggregates shall at all times be such as to eliminate segregation or contamination of the various sizes and to prevent contamination of materials by dust. Stockpiles shall be kept flat and the formation of high cone-shaped piles shall not be permitted. When conveyor belts are used for stockpiling aggregates, the Engineer may require the use of baffle-chutes or perforated chimneys.

4 Where trucks are used to construct stockpiles, the stockpiles shall be constructed one layer at a time with trucks depositing their loads as close to the previous load as possible. The use of tractors or loaders to push material deposited at one location to another location in the stockpile shall not be allowed during the construction of the stockpile, and their use shall be limited to levelling the deposited material only.

(a)

iii. The Contractor shall submit the name(s) of personnel responsible for receipt, inspection, and record keeping of PG binder materials. Contractor plant personnel shall document specific storage tank(s) where binder will be transferred and stored until used, and provide binder samples to the Engineer upon request. The person(s) shall assure that each shipment (tanker truck) is accompanied by a statement certifying that the transport vehicle was inspected before loading and was found acceptable for the material shipped and that the binder will be free of contamination from any residual material, along with 2 copies of the bill of lading.

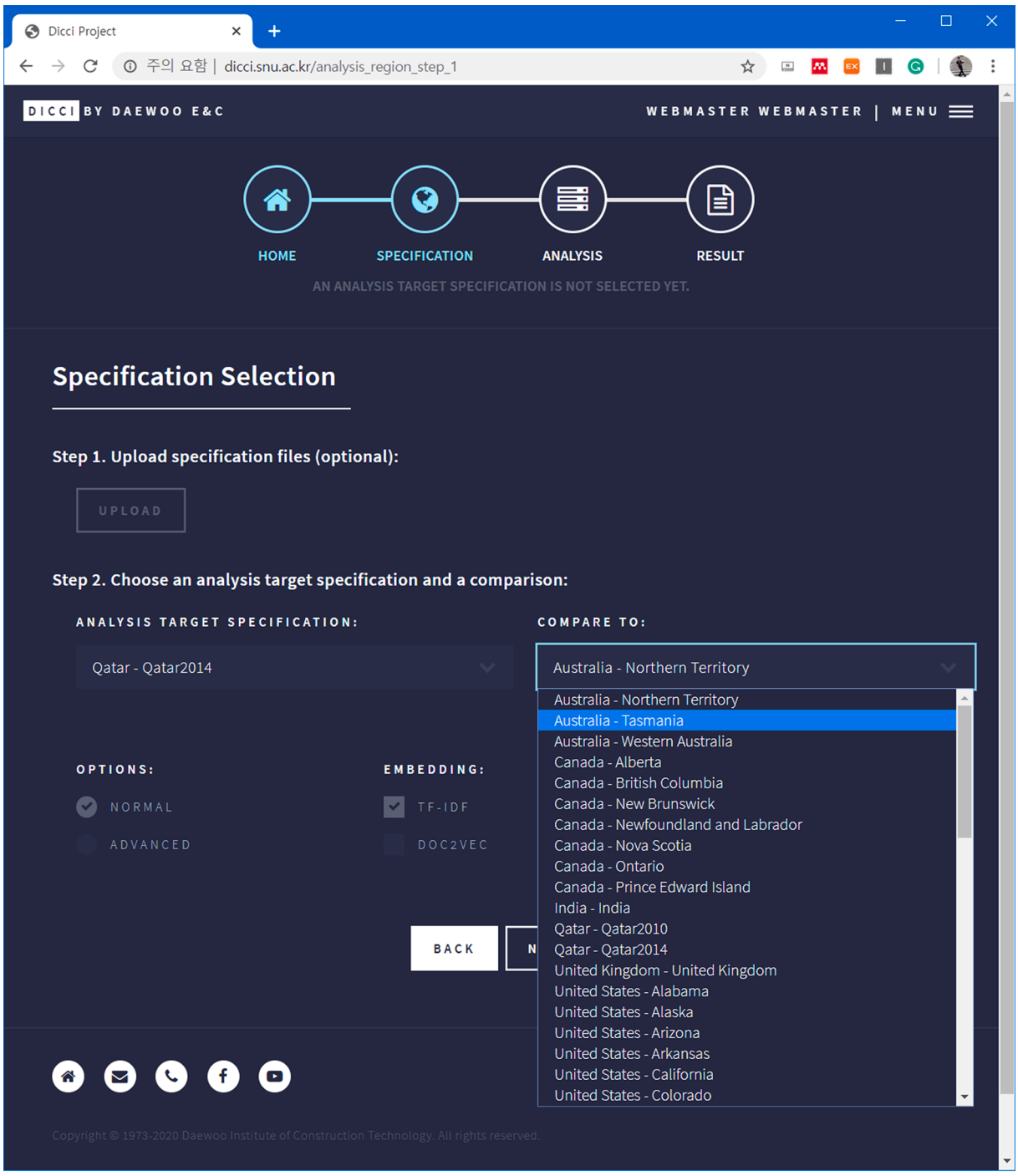
(b)

Fig. 9. Example result of relevant provision pairing: (a) 'Qatar_2014_06_05,' (b) 'United States_Connecticut_2018_19_04.'

5. Evaluation and discussions

5.1. Case study for evaluation

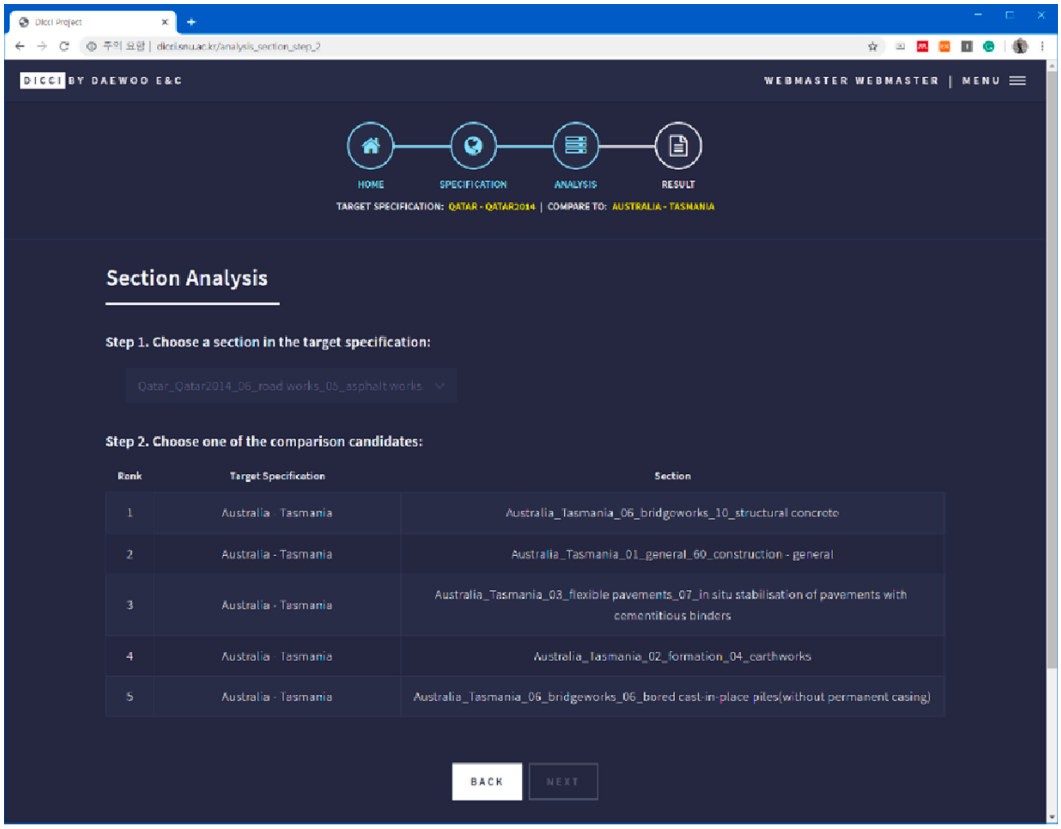
The authors conducted experiments to validate the practical usefulness of the prototype by asking construction practitioners to review a part of the Qatar Construction Specification in 2014. Excluding 17 appendix provisions, the experiments used 58 provisions out of 75 from 'Qatar_2014_06_05.' A total of 12 construction practitioners (i.e., participant IDs of 3, and 15 to 25 in Table 3) were involved in the experiments. The average work experience of the participants was six years. Although the participants included a number of researchers who do not have much field experience, the experiment is still reasonable since, in many cases, construction practitioners are required to review specifications early on in their careers when they have under five years of experience; in other words, the researchers were able to represent novice field engineers.



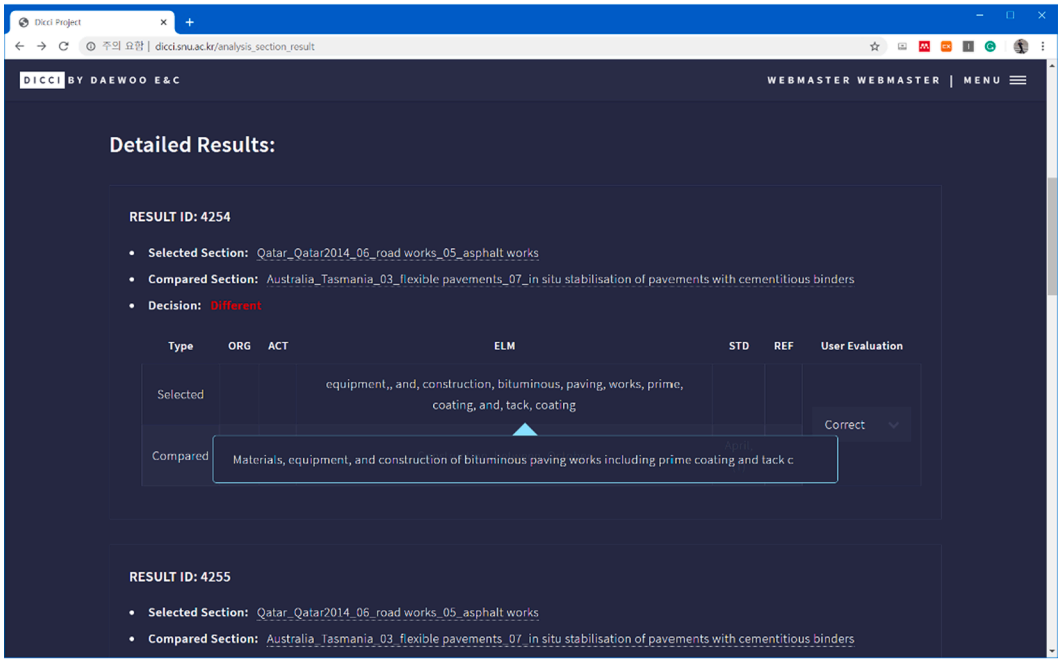
(a)

Fig. 10. Specification review process of the web-based prototype: (a) data selection, (b) provision recommendation, (c) keyword detection, (d) report exportation.

The experiments attempted to demonstrate the performance of the proposed model in two cases; reviewing a very similar document and reviewing a totally different document. Therefore, the authors provided 'Qatar_2010_06_05' and 'United States_Connecticut_2018_19_04' as relevant specifications for the participants. 'Qatar_2010_06_05′ showed very similar semantic properties to 'Qatar_2014_06_05,' whereas 'United States_Connecticut_2018_19_04' was noticeably different. The authors conducted a preliminary study by reviewing 'Qatar_2014_06_05' and 'United States_Connecticut_2018_19_04,' and concluded that only 25 of the 58 provisions of 'Qatar_2014_06_05' had relevant provisions in 'United States_Connecticut_2018_19_04.' The participants were randomly divided into two groups: the control group and the experiment group. The control group reviewed the provisions manually, while the experiment group reviewed the same provisions using the prototype.



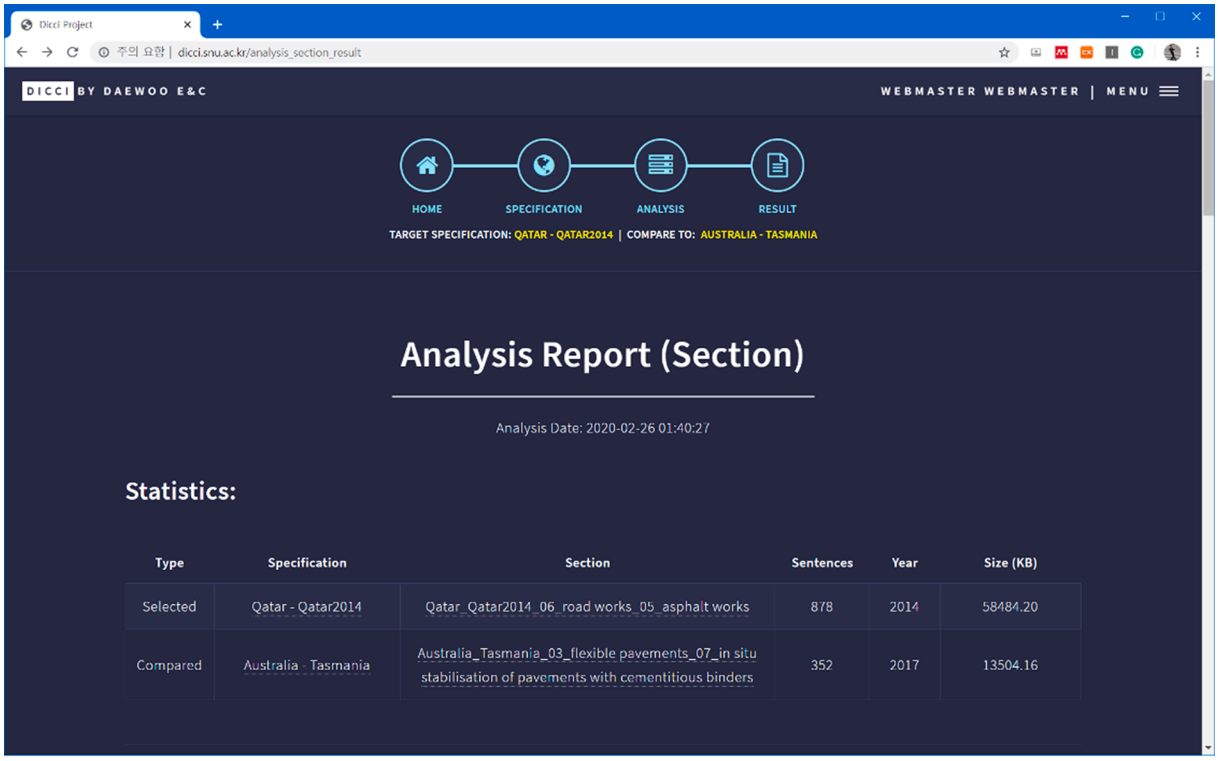
(b)



(c)

Fig. 10. (*continued*).

The case study experiments included several assumptions. First, the proposed model understands the contents of construction specifications based on the six information types (i.e., ORG, ACT, ELM, STD, REF, and NON) determined in the research. Second, the proposed model analyzes the specifications based on the scale of provision—a set of continuous sentences that share the lowest level of heading. Third, the authors set a



(d)

Fig. 10. (*continued*).

ground truth value using the review results of the participant with the longest work experience (i.e., participant ID of 3) to demonstrate the feature of the proposed model that narrows the gap between beginners and experts. These assumptions have opportunities for further research, which is discussed in the conclusion.

5.2. Experimental results of case study

The experiments collected 17 results; nine participants reviewed 'Qatar_2014_06_05' using 'Qatar_2010_06_05' and the remaining eight participants used 'United States_Connecticut_2018_19_04.' The authors estimated (1) the time spent for the review process and (2) the review performance of each experiment, and evaluated the values with the ground truth.

The results of the control group that analyzed 'Qatar_2014_06_05' and 'Qatar_2010_06_05' were estimated as shown in Table 11. The control group spent 62.75 min reviewing the 58 provisions and returned disappointing performances, with the average precision, recall, and F1 scores of 0.619, 0.557, and 0.586, respectively. Meanwhile, the experiment group spent only 48.75 min (i.e., a reduction of 22.3%) to review the same provisions, and their performance was relatively accurate, with an average precision, recall, and F1 scores of 0.755, 0.705, and 0.728, respectively (i.e., an increase 24.2% in the F1 score) (Table 12).

Table 11
Review results of the control group ('Qatar_2014_06_05' and 'Qatar_2010_06_05').

Participant ID	Duration (minutes)	Precision	Recall	F1 Score
18	60	0.623	0.576	0.598
19	66	0.717	0.652	0.683
20	73	0.500	0.470	0.484
23	60	0.636	0.530	0.579
Average	62.75	0.619	0.557	0.586

Table 12
Review results of the experiment group ('Qatar_2014_06_05' and 'Qatar_2010_06_05').

Participant ID	Duration (minutes)	Precision	Recall	F1 Score
22	63	0.685	0.561	0.617
15	48	0.818	0.818	0.818
17	44	0.800	0.788	0.794
24	40	0.717	0.652	0.683
Average	48.75	0.755	0.705	0.728

The results of the control group that analyzed 'Qatar_2014_06_05' and 'United States_Connecticut_2018_19_04' were estimated as shown in Table 13. They spent 124.33 min reviewing the given documents, and the average precision, recall, and F1 scores were 0.521, 0.563, 0.541, respectively. The results of the experiment group were estimated as shown in Table 14. The automated review program assisted participants in reducing their working hours considerably, taking only 39.33 min on average (i.e., a reduction of 68.4%). Although the accuracy showed similar performances to those of the control group, with the average precision, recall, and F1 scores of 0.531, 0.540, and 0.535, respectively, the reduced time might compensate for the lack of performance

Table 13
Review results of the control group ('Qatar_2014_06_05' and 'United States_Connecticut_2018_19_04').

Participant ID	Duration (minutes)	Precision	Recall	F1 Score
17	196	0.629	0.672	0.650
19	99	0.441	0.448	0.444
21	78	0.493	0.569	0.528
Average	124.33	0.521	0.563	0.541

Table 14
Review results of the experiment group ('Qatar_2014_06_05' and 'United States_Connecticut_2018_19_04').

Participant ID	Duration (minutes)	Precision	Recall	F1 Score
16	33	0.500	0.500	0.500
18	50	0.541	0.569	0.555
25	35	0.552	0.552	0.552
Average	39.33	0.531	0.540	0.535

improvement. In addition, participants' performance variation of the experiment group also decreased, which indicates the automated review results are more consistent than the manual results.

5.3. Discussions

The prime beneficiaries of the proposed system are field engineers who have less experience in reviewing specifications and contractors who lack the time and human resources to review such large and complex documents. To address the industrial necessities, the authors considered the experimental results in terms of time efficiency, accuracy of detecting erroneous provisions, and robustness to reviewers' subjectivity.

First, the prototype achieved considerable time efficiency. According to the experimental results of 'Qatar_2014_06_05' and 'Qatar_2010_06_05,' the manual review process required more than 20% of working hours compared to reviewers supported by the automated method. In addition, according to the experimental results of 'Qatar_2014_06_05' and 'United States_Connecticut_2018_19_04,' the automated construction specification review can easily bypass non-relevant provisions.

Second, the proposed method provided higher accuracy on pairing the relevant provisions and determining the differences between each pair. This process facilitates identifying erroneous provisions that are not appropriate to the site conditions. In the experimental results of 'Qatar_2014_06_05' and 'Qatar_2010_06_05,' less experienced participants showed disappointing performances in their review results compared to the most experienced expert when conducting the review manually (i.e., an F1 score of 0.586). However, the experiment group acquired an F1 score of 0.728 for the correct answers. These results indicate that unskilled engineers can perform at a similar level to that of their more experienced counterparts (with an F1 score of 0.728) with the support of the proposed method. Although the result of the most experienced professional might contain errors and mistakes, the experiments demonstrated that the proposed method could narrow the gap between practitioners. In addition, although most of the provisions from 'Qatar_2014_06_05' and 'United States_Connecticut_2018_19_04' were not relevant, users of the automated process can easily bypass non-relevant provisions. Reducing the required time to review specifications would allow contractors to focus more on the risky provisions despite the bidding process's tight schedule.

Lastly, the prototype demonstrated that it avoids reviewers' subjectivity. The manual review produced conflicting results among reviewers due to the different levels of experience and capability. For example, the reviewers returned incompatible results for a provision (ID '5.1.4.4') in 'Qatar_2014_06_05': one participant answered no relevant provision, while another answered provision '5.1.3.3' in 'Qatar_2010_06_05' should be paired. Conversely, the proposed method can provide consistent results and ensure the review is robust to subjectivity. Consistency is crucial in risk management as review errors can be predictable; further improvement direction can also be determined based on errors.

In conclusion, the proposed method can contribute to the construction specification review by shortening the processing time, narrowing the gap between beginners and experts, and reducing the subjectivity of the results. The most important findings of this research are that the proposed method successfully analyzed the varied semantic properties between different construction specifications and automatically provided a draft of the review results to the reviewers. However, the accuracy of detecting erroneous provisions should be improved for practical usage. The automated construction specification review can be improved by ensuring the model learns large amounts of data from various documents, considering additional text features such as provision categories, and applying state-of-the-art deep learning architectures.

6. Conclusion

Beyond several existing approaches for automated document reviews in the construction industry, the specification review process requires individuals to analyze different documents each time due to their varied semantic properties. Varied semantic properties across different specifications disrupt existing approaches to automating the review process. The objective of this research was to propose an automated system for reviewing construction specifications by analyzing the varied semantic properties: different vocabulary, different sentence structures, and differently organized provisions. The authors proposed a semantic construction thesaurus to assist the computer in understanding the text irrespective of the vocabulary used in a document. Based on the thesaurus, the NER model was developed to facilitate the computer to understand the text irrespective of different sentence structures. The authors also developed a provision-pairing model to facilitate the computer to understand the text irrespective of differently organized provisions. Finally, the UI and UX of the web-based prototype was designed for the easy use of industry practitioners. The experimental results demonstrate that the research can address the semantic conflicts across different documents and facilitate an automated review process of construction specifications with a sufficient level of industrial applicability. The results show that study participants were able to review a specification against a different one with much less time and effort needed to analyze and understand the document.

The experimental results of the prototype demonstrate the necessity and practical usefulness of the proposed method for automatic specification review in the construction industry. The authors suggest several industrial application scenarios of the research. First, the proposed method will enable contractors to review specifications in the early phases of the construction project, which will improve the risk management process. Once provided with a construction specification from the client, the contractor would input the document into the prototype and select a standard specification to compare it. The prototype would then analyze every provision of the given specification. If erroneous provisions are detected, the client and contractor can discuss them in detail to correct the provisions and reduce project risk. Second, the proposed method can be used repeatedly throughout the construction project. Since construction projects commonly last years and the site conditions can change during the project, field engineers should frequently review and analyze the requirements. Occasionally, the client might provide a new construction specification with modified or updated clauses. As the program automatically identifies the most relevant provision from other specifications, the field engineers can easily find other provisions that are relevant to the construction specification. Third, the client responsible for drafting the project requirements in construction specifications can also benefit from the proposed method. The method will allow clients to describe the requirements based on the

relevant provisions of other specifications with much less effort needed to search for these provisions. The proposed method also facilitates a preliminary review of the construction specification.

This research provides several contributions to the body of knowledge in the construction industry. First, the authors identified and addressed the three types of semantic conflicts in construction specifications that cause difficulties in automating the review process. The different vocabulary, different sentence structures, and differently organized provisions require additional tasks for the automated review process in practice. Addressing these problems with state-of-the-art NLP techniques, this research gets closer to achieving a fully automated construction specification review method. Second, the proposed method can be applied to other domains, for example, other languages, other types of documents, and other industries. Since the research utilizes a distributed language model, the proposed method can handle other data if sufficient training data is provided and several hyperparameters are customized. Therefore, this research can contribute to risk management and mitigation in the construction industry, particularly related to various documents, including contract documents, non-conformance reports, accident reports, and inspection reports. Lastly, the results can support risk management in construction projects, especially given that manually reviewing construction specifications is difficult because of the tight schedule of the bidding process, the insufficient number of available professionals, and the large volume of information (over several thousand pages) to be reviewed. Using the proposed automated method, contractors can automatically check whether the requirements meet the local conditions and are consistent throughout the entire specification at the bidding stage. In doing so, they can review the provisions more efficiently and find erroneous provisions with different standard criteria, which will facilitate proactively preparing for potential problems.

The authors suggest several areas of interest for future studies investigating the fully automated review of construction specifications according to the research contributions. First, the assumptions of the case study suggest opportunities for further research. Follow-up studies can improve the results by detecting requirement keywords more accurately, considering various units of text (e.g., various levels of provisions), and building a precise ground truth for specification reviews (i.e., rather than just using the results of the participant with the longest work experience). Second, since most documents in the construction industry are managed in PDF format, the document data should be reformed into TXT format for the language models. Therefore, better PDF-to-TXT conversion techniques would facilitate text analysis in the construction industry. In addition, a provision searching system would facilitate the usage of automated document reviews. Beyond pairing the relevant provisions and detecting different requirements, a fully automated system should also be able to suggest a correct provision for the user in advance.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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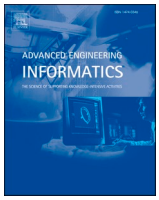
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Corrigendum



Corrigendum to "Automated system for construction specification review using natural language processing" [Adv. Eng. Inf. 51 (2022) 101495]

Seonghyeon Moon [a,b], Gitaek Lee [a], Seokho Chi [a,b,*]

[a] Department of Civil and Environmental Engineering, Seoul National University, Seoul 08826, Republic of Korea
[b] Institute of Construction and Environmental Engineering, Seoul National University, Seoul 08826, Republic of Korea

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* Corresponding author at: Department of Civil and Environmental Engineering, Seoul National University, Seoul 08826, Republic of Korea.
E-mail addresses: blank54@snu.ac.kr (S. Moon), lgt0427@snu.ac.kr (G. Lee), shchi@snu.ac.kr (S. Chi).

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